

Analysis design and reference mapping

Methods adapted to mouse iNKT; biological conclusions are derived from these data.

Analysis	Reference	Input / definition	Implementation
Cytotoxicity	PAGER Fig. 5C/D	Core: Prf1, Gzma, Gzmb; E1: +Ctla2a	Mean log1p expression; matched-control score sensitivity
GO cluster identity	GAFA Fig. 4c	Cluster vs rest within each tissue	Wilcoxon, gene BH <= .05 and log2FC >= .25
GO treatment response	GAFA functional comparison	T2-up and Ctrl-up genes tested separately	Hypergeometric; full eligible GO BP/MF/CC families
Local trajectories	PAGER Fig. 5D	Recompute diffusion map on tissue-specific graphs	Precursor-rooted DPT; 20 shared bins; 3-root sensitivity
Robustness	Adaptation to this dataset	Stable clusters; five QC-matched resamples	Effects and direction consistency; no animal replication

GO uses official mouse annotations. An enrichment of upregulated genes does not prove pathway activation.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.